NUMBER GUSSING GAME

Mini Project Report

Project Title: NUMBER GUSSING GAME

Programming Language: Python

IDE: Visual Studio Code

Author: SRINIDHI U

1. Objective

The main objectives of this project are:

1. To develop a simple Number Guessing Game using Python.
2. To generate a random number between 1 and 100 using the `random` module.
3. To accept number input from the user.
4. To compare the user's guess with the computer-generated number.
5. To display hints using conditional statements.
6. To repeat the guessing process using a `while` loop.
7. To handle invalid inputs using exception handling.
8. To improve logical thinking and problem-solving skills.

2. Abstract

The Number Guessing Game is a simple and interactive mini project developed using the Python programming language.

In this game, the computer randomly selects a secret number between 1 and 100. The user must guess the correct number.

After each guess, the program provides hints to help the user:

- If the guessed number is too low, it displays "That's too low!"
- If the guessed number is too high, it displays "That's too high!"
- If the guessed number is correct, it displays "Congratulations! You guessed the number!"

The program also handles invalid inputs using exception handling. If the user enters an invalid value, the program displays an appropriate error message.

This project helps beginners understand the basic concepts of Python programming.

3. Software Requirements

Hardware Requirements

- Computer or Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

Software Requirements

- Operating System: Windows
- Programming Language: Python 3
- Code Editor: Visual Studio Code
- Python Libraries: `random` (built-in module)

4. EXCEPTION HANDLING:

```
except ValueError:
    print("Please enter a valid number between 1 and 100.")
```

5. Algorithm

Step 1: Start the program.

Step 2: Import the `random` module.

Step 3: Generate a random number between 1 and 100 using `random.randint(1, 100)`.

Step 4: Display the game instructions to the user.

Step 5: Use a `while` loop to repeatedly ask the user to guess the number.

Step 6: Accept the user's input and convert it into an integer.

Step 7: Check whether the entered number is between 1 and 100.

Step 8: If the entered number is outside the allowed range, raise a `ValueError` and display an error message.

Step 9: If the guessed number is less than the secret number, display "That's too low!"

Step 10: If the guessed number is greater than the secret number, display "That's too high!"

Step 11: If the guessed number matches the secret number, display "Congratulations! You guessed the number!"

Step 12: Use the `break` statement to stop the loop after the correct guess.

Step 13: Handle invalid input using the `try-except` block.

Step 14: End the program.

6. Source Code

```
import random

secret_number = random.randint(1, 100)

print("Welcome to the Number Guessing Game!")
print("Guess a number between 1 and 100.")

while True:
    try:
        guess = int(input("Enter your guess: "))

        if guess < 1 or guess > 100:
            raise ValueError

        if guess < secret_number:
            print("That's too low!")

        elif guess > secret_number:
            print("That's too high!")

        else:
            print("Congratulations! You guessed the number!")
            break

    except ValueError:
        print("Please enter a valid number between 1 and 100.")
```

---

7. SamplE OUTPUT:

```
Welcome to the Number Guessing Game!
Guess a number between 1 and 100.
Enter your guess: 25
That's too low!

Enter your guess: 85
That's too high!

Enter your guess: 50
Congratulations! You guessed the number!
```

8. Advantages

- Simple and easy to understand.
- Improves logical thinking.
- Demonstrates random number generation.
- Provides interactive user input.
- Uses loops and conditional statements.
- Handles invalid input using exception handling.
- Suitable for Python beginners.

9. Future Enhancements

1. Limit the number of attempts.
2. Display the total number of guesses.
3. Add difficulty levels such as Easy, Medium, and Hard.
4. Allow the user to play again.
5. Add a graphical user interface (GUI).
6. Maintain a score based on the number of attempts.

10. Conclusion

The Number Guessing Game is a beginner-friendly Python mini project that demonstrates the practical implementation of fundamental programming concepts.

The project uses the `random` module, `while` loop, conditional statements, user input, and exception handling.

By developing this project, beginners can improve their understanding of Python programming, logical reasoning, and error management.